$$n = 100$$

$$\lambda = np = 4$$

(i) $P(X=1) = \frac{e^{-\lambda}\lambda}{1!} = e^{-4}(4) = 0.0183 \times 4$

$$= 0.0732$$

(ii) $P(X=0) = \frac{e^{-\lambda}\lambda^0}{0!} = e^{-4} = 0.0183$

Number of boxes having no defective pin = 200×0.0183

= 3.660

= 4

10.2 Continuous distributions:

10.2.1 Rectangular or Uniform Distribution

Definition

A random variable X is said to have a continuous Uniform distribution over the interval (a, b) if its probability density function is

$$f(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & otherwise \end{cases}$$

Characteristics of Uniform Distribution

(i) a and b are the parameters of the Uniform distribution and we write $X \sim U(a, b)$

(ii) The distribution is also known as Rectangular distribution, as the curve

(iii) $y = f(x)$ describes a rectangle over the x-axis and between ordinates at x = a and $x = b$.

(iv) f $X \sim U(-a, a)$ then its p.d.f. is $f(x) = \begin{cases} \frac{1}{2a} & -a < x < a \\ 0 & otherwise \end{cases}$

Constants of uniform distribution X ~ U (a, b)

(i) Mean $\mu = \frac{a+b}{2}$

(ii) Variance $\sigma^2 = \frac{(b-a)^2}{12}$

(iii) Median $= \frac{a+b}{2}$

(iv) Skewness = 0

(v) Kurtosis = $-\frac{6}{5}$

(vi) $Q_1 = \frac{3a+b}{4}$ (vii) $Q_3 = \frac{a+3b}{4}$